

4. SYSTEM ANALYSIS AND CLASSIFICATION

Experiment 4.1

Discrete -Time System

PROGRAM

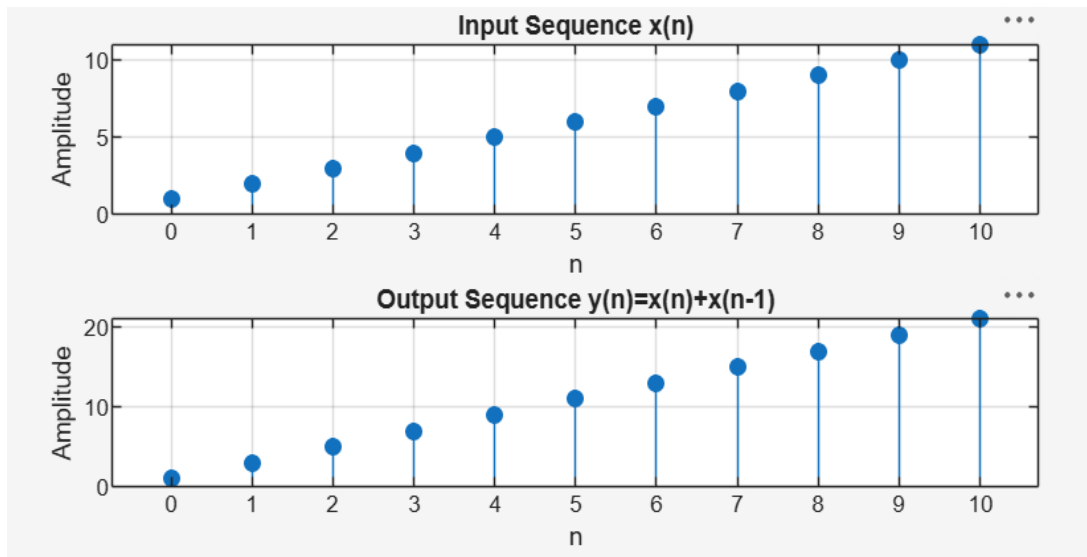
```
clc;
clear;
close all;
% Input sequence
n = 0:10;
x = [1 2 3 4 5 6 7 8 9 10 11];
% Discrete-Time System
y = filter([1 1],1,x);
% Display output
disp('Output Sequence:');
disp(y);
% Plotting
figure;
subplot(2,1,1);
stem(n,x,'filled');
grid on;
xlabel('n');
ylabel('Amplitude');
subplot(2,1,2);
stem(n,y,'filled');
grid on;
title('Output Sequence  $y(n)=x(n)+x(n-1)$ ');
xlabel('n');
ylabel('Amplitude');
```

```

Command Window
Output Sequence:
    1    3    5    7    9   11   13   15   17   19   21

>>

```



Experiment 4.2

Causal/Non-Causal System

```

clc;

clear;

close all;

% Input sequence

n = 0:10;

x = [1 2 3 4 5 6 7 8 9 10 11];

% Causal System: y(n)=x(n)+x(n-1)

y_causal = filter([1 1],1,x);

% Non-Causal System: y(n)=x(n)+x(n+1)

y_noncausal = x + [x(2:end) 0];

% Plotting

figure

```

```

subplot(3,1,1);

stem(n,x,'filled');

grid on;

title('Input Sequence x(n)');

xlabel('n');

ylabel('Amplitude');

subplot(3,1,2);

stem(n,y_causal,'filled');

grid on;

title('Causal System Output y(n)=x(n)+x(n-1)');

xlabel('n');

ylabel('Amplitude');

subplot(3,1,3);

stem(n,y_noncausal,'filled');

grid on;

title('Non-Causal System Output y(n)=x(n)+x(n+1)');

xlabel('n');

ylabel('Amplitude');

```

